UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

Course Syllabus

1	**Course Title**	Introduction to Computer Systems
2	**Course Code**	CSE 1110
3	**Trimester and Year**	Fall 2026
4	**Pre-requisites**	
5	**Credit Hours**	1.0
6	**Section**	
7	**Class Hours**	
8	**Classroom**	
9	**Instructor's Name**	Asif Ahmed Utsa
10	**Email**	asif@cse.uiu.ac.bd Mobile: 01700545293
11	**Office**	Room # 419 (A)
12	**Counselling Hours**	
13	**Text Book**	Programming in ANSI C
14	**Reference**	Lecture documents
15	**Course Contents (approved by UGC)**	
16	**Course Outcomes (COs)**	
17	**Teaching Methods**	Lecture, laboratory exercise and reports.

CO	Description
CO1	Identify the components of a computer and demonstrate its internal mechanisms.
CO2	Derive the solution steps and flowchart of any programs.
CO3	Analyze the code to find errors and trace outputs manually.
CO4	Use a programming language to write error-free programs.

18 CO with Assessment Methods

CO	Assessment Method	(%)
-	Attendance	5
-	Class Performance	30
-	Assignment	15
-	Viva	10
CO1, CO2	Midterm	20
CO3,CO4	Final	20

19 Mapping of COs and Program outcomes

CO	Statement	Bloom's Domain
CO1	Identify the components of a computer and demonstrate its internal mechanisms.	**C**
CO2	Derive the solution steps and flowchart of any programs.	**C**
CO3	Analyze the code to find errors and trace outputs manually.	**C**
CO4	Use a programming language to write error-free programs.	**C**

20 Lab Outline

Class	Lecture/Activities	Evaluation Methods	COs
1	**UIU Rules** **Code.org** **Internal mechanism of computer** Parts of a computer and their basic functionalities, how computer works, memory vs. storage, Applications of different software's	- **Group discussion**	**CO1**
2	**Algorithm and Flowcharts** Write algorithms and flowcharts for various real-life problems.	- **Group discussion**	**CO2**
3	**Introduction to C and Beecrowd** Codeblocks, basic syntax, data types, variables, built-in functions, account create, solving problems	-	**CO3, CO4**
4	**Problem Solving** Arithmetic operator, Logical and Relational operator, solve real- life problems	- **Group discussion** - **Class Test 1**	**CO3, CO4**
5	**Conditional Statement** Basic If_else_if_else, solve real life problems	- **Group discussion**	**CO3, CO4**
6	**MID-TERM EXAM**		
7	**Conditional Statement Review** conditional statement, solve real-life problems, Introduction to switch case,	- **Group discussion** - **Exercise problem**	**CO3, CO4**
8	**Conditional Statement** Nested switch case and solve real- life problems	- **Group discussion** - **Exercise problem**	**CO3, CO4**

	9	**Problem Solving and Loop Introduction** Introduction to Loop	- **Class Test 2**	**CO3, CO4**
	10	**Loop or Iterative Approach** Loop, use case, different types of loops, for and while loop, syntax, solve real-life problems	- **Group discussion** - **Exercise problem** - **Assignment - 2**	**CO3, CO4**
	11	**Problem Solving** Problem solving, discussion about final topics and marks distribution	- **Class Test 3**	**CO3, CO4**
	12	**Final EXAM and Viva**		

Appendix 1: Assessment Methods

Assessment Types	**Marks**
Attendance	5%
Class Performance	30%
Assignment	15%
Viva	10%
Mid Term	20%
Final Exam	20%

Appendix 2: Grading Policy

Letter Grade	**Marks %**	**Grade Point**	**Letter Grade**	**Marks%**	**Grade Point**
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

	Program Outcomes
1	**Engineering knowledge:** Apply knowledge of mathematics, natural science, engineering fundamentals and Computer Science and Engineering to the solution of complex engineering problems.
2	**Problem analysis:** Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

3	**Design/development of solutions:** Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4	**Investigation:** Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions
5	**Modern tool usage:** Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an

	understanding of the limitations.
6	**The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.
7	**Environment and sustainability:** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.
8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
9	**Individual work and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**Project management and finance:** Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's 13 own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.